



Batteries

Will someone please give the Energizer Bunny a run for its money?

Ram Mohan Rao

We've talked about human nature elsewhere in this issue, and we return to it now: how many times have you said to someone on the phone, "I'll call you back! My battery is runnin'..." [Beep Beep Beep] And then, of course, how many times have you gone asking people around the office, "Does anyone have a <insert brand name here> charger?" All this is as much a part of our lives as our mobility.

Is There An Answer?

The answer is simple (except in cases where you don't have access to a power outlet): always carry your charger with you in the pocket of your jeans (or in your purse). And charge your device religiously every morning—all it takes is a half-hour each day, and it can happen in the background while, for example, you get ready for work.

Still, we hanker for longer battery life! We even curse the phone or PDA or laptop—as in "This bloody thing keeps running out!"

Well, run out it will, with current-gen battery technologies. So, being pragmatic about it, we (a) realise we won't change our ways, and (b) hope for all the innovative companies out there to come out with longer-lasting batteries.

Some companies, knowing how demanding and lazy we are, realised there's a zillion-dollar market for "betteries." It turns out that the same technology that can be used in hybrid cars—as in fuel cells—can also be used for portable devices.

The Fuel Cell

"Fuel cells for portables," as a phrase, was being bandied about quite a long time ago. In 2003 or so, we'd chanced upon a prototype laptop-fuel cell combo. The fuel cell looked like a little tanker attached to the back of the laptop; it was the same width as the laptop. It was an ugly prototype that no-one would use. But we

now have fuel cells for laptops and other portables that are much smaller than the devices they power.

Fuel cells are mainly different (from the user's perspective) from regular batteries in that you don't connect them to a power outlet to recharge them; they're cartridges, and you refill them by actually pouring fuel (often methanol) into them. That's if the companies decide to market them as refillable.

The Story Thus Far

At CES 2006, Panasonic showed off a fuel cell-equipped laptop. The company said it would provide 20 hours on a single 200 cc tank of methanol, and the fuel would be provided in cartridges.

15 May 2006, Casio hopped on the portable-fuel-cell bandwagon, promising to offer a compact battery for laptops in 2007. The prototype they had was a 6.5 x 1.9 x 1.8 cm unit, and its power capacity, translated into the size of today's batteries, meant a fuel cell the size of a standard Li-ion laptop battery would last four times as long, according to Casio—close to Panasonic's 20 hours. Temperature is a potential problem: 280°C is the temperature the cell gets to during operation! However, the cell packaging means the external surface temperature would be more like 40 degrees—certainly bearable.

We learnt that it's Toshiba that has the distinction of having produced the world's smallest methanol fuel cell. Back in September 2005, the company also developed two prototypes of fuel-cell-powered MP3 players, Flash-based and hard disk-based respectively. The Flash player ran 35 hours on a single 3.5 ml charge of concentrated methanol, and the hard disk-based player runs about 60 hours on a single 10 ml charge—if you use it for five hours a day, it will last 12 days!

The above list of prototypes is in no way exhaustive: there are many companies out there developing micro fuel cells. But what's nanotech doing? Well, no prototype gadgets yet, but nationalgeo-

graphic.com reported the following back in January 2005:

"Scientists have invented a plastic solar cell that can turn the sun's power into electrical energy... The plastic material uses nanotechnology and contains the first solar cells able to harness the sun's infrared rays... Like paint, the composite can be sprayed onto other materials and used as portable electricity. A sweater coated in the material could power a cell phone or other wireless devices."

That "could power" doesn't sound convincing enough, somehow: besides, we haven't heard anything about nano-sprayed fuel since that press release. Perhaps 2010 is a realistic estimate for prototypes based on that technology.

But coming back to fuel cell-powered gadgets, they're very much there, except they aren't on store shelves yet. However, 2007 could well be the year.

Not Everything Is Rosy

For once, we end on a negative note: will 2007 really be the year? "Toshiba May Bid Laptop Batteries Goodbye: New fuel cell, coming next year, could provide 10 hours of power for portable PCs." That was what IDG News Service reported in March of 2003. Where's that fuel cell?

Also, once these gadgets are out, we're going to complain about (a) size; (b) having to carry bottles of fuel; and (c) the fuel spilling into our pockets or handbags. But that's just human nature! ■

ram_mohan@thinkdigit.com

